

Baseline Sentiment Model Report

Aspect-Based Sentiment Analysis on Laptop Reviews (Flipkart) — Week 07

Problem Definition

This phase builds a baseline machine learning model as a first step toward the project's Aspect-Based Sentiment Analysis goal. Before extracting sentiment at the level of individual product aspects (battery, display, build quality, etc.), a simpler overall sentiment classification model was built to establish a working pipeline and a genuine performance baseline.

Target variable: A binary sentiment label derived from star rating: ratings 1–3 mapped to Negative, ratings 4–5 mapped to Positive.

Why 2-class instead of 3-class (with Neutral): The dataset is heavily skewed toward 5-star ratings (58%, per Week 05 EDA), so a 3-class split would leave very few examples for a Neutral class, producing an unstable baseline. A 2-class split provides a simpler, more reliable starting point. This decision will be revisited with the mentor, and can be changed once this baseline pipeline is confirmed to work end-to-end.

Features: The review text (`review` column), converted into numeric features using TF-IDF (5,000 features, English stop words removed).

Data Split

The dataset (16,991 reviews) was split into training (80%, 13,592 reviews) and test (20%, 3,399 reviews) sets using a stratified split, preserving the original class balance (81.6% Positive, 18.4% Negative) in both sets. TF-IDF was fit only on the training set and then applied to the test set, to prevent data leakage between the two.

Baseline Model

A majority-class baseline (DummyClassifier) was established before training a real model. This baseline ignores review text entirely and always predicts the majority class (Positive), providing an honest floor that a genuinely useful model needs to beat.

Class	Precision	Recall	F1-score
Negative	0.00	0.00	0.00
Positive	0.82	1.00	0.90
Accuracy			0.82

The baseline achieved 82% accuracy but 0% recall on the Negative class, meaning it failed to identify a single negative review. This demonstrates that accuracy alone is a misleading metric on this imbalanced dataset, and establishes the real bar the actual model needs to clear: catching negative sentiment, not just overall accuracy.

Baseline ML Model: Logistic Regression

A Logistic Regression model was trained on the TF-IDF features, using `class_weight="balanced"` to address the class imbalance and prevent the model from defaulting to the majority class the way the dummy baseline did.

Class	Precision	Recall	F1-score
Negative	0.60	0.80	0.69
Positive	0.95	0.88	0.91
Accuracy			0.87

Baseline vs Model Comparison

Metric	Baseline	Logistic Regression
Accuracy	82%	87%
Negative Recall	0%	80%
Negative Precision	0%	60%
Negative F1-score	0.00	0.69

Confusion Matrix

	Predicted Negative	Predicted Positive
Actual Negative	497	128
Actual Positive	328	2446

The model correctly identifies 497 of 625 actual Negative reviews (79.5% recall), while missing 128. It also produces 328 false positives (Positive reviews incorrectly flagged as Negative), explaining the 60% precision on the Negative class. For a first baseline, this is a reasonable trade-off: the model prioritizes catching negative sentiment at some cost of false alarms, which aligns with the project's goal of identifying dissatisfied customers.

Conclusion

A baseline sentiment classification model was built as a first step toward this project's Aspect-Based Sentiment Analysis goal. Star ratings were mapped into a binary sentiment label, and review text was converted into numeric features using TF-IDF, with the vectorizer fit only on training data to prevent leakage.

A majority-class baseline achieved 82% accuracy but completely failed to identify any negative reviews (0% recall), demonstrating that accuracy alone is a misleading metric on this imbalanced dataset. A Logistic Regression model trained on the TF-IDF features improved overall accuracy to 87%, and more importantly, achieved 80% recall on the Negative class, correctly identifying most negative reviews.

This confirms that review text contains a genuine, learnable signal for sentiment, and establishes a working baseline pipeline (text to features to prediction to evaluation) that can be extended in future work, particularly toward per-aspect sentiment extraction, the next phase of this project.

Next steps: discuss the 2-class versus 3-class (with Neutral) decision with the mentor, and begin scoping the Aspect Extraction phase.